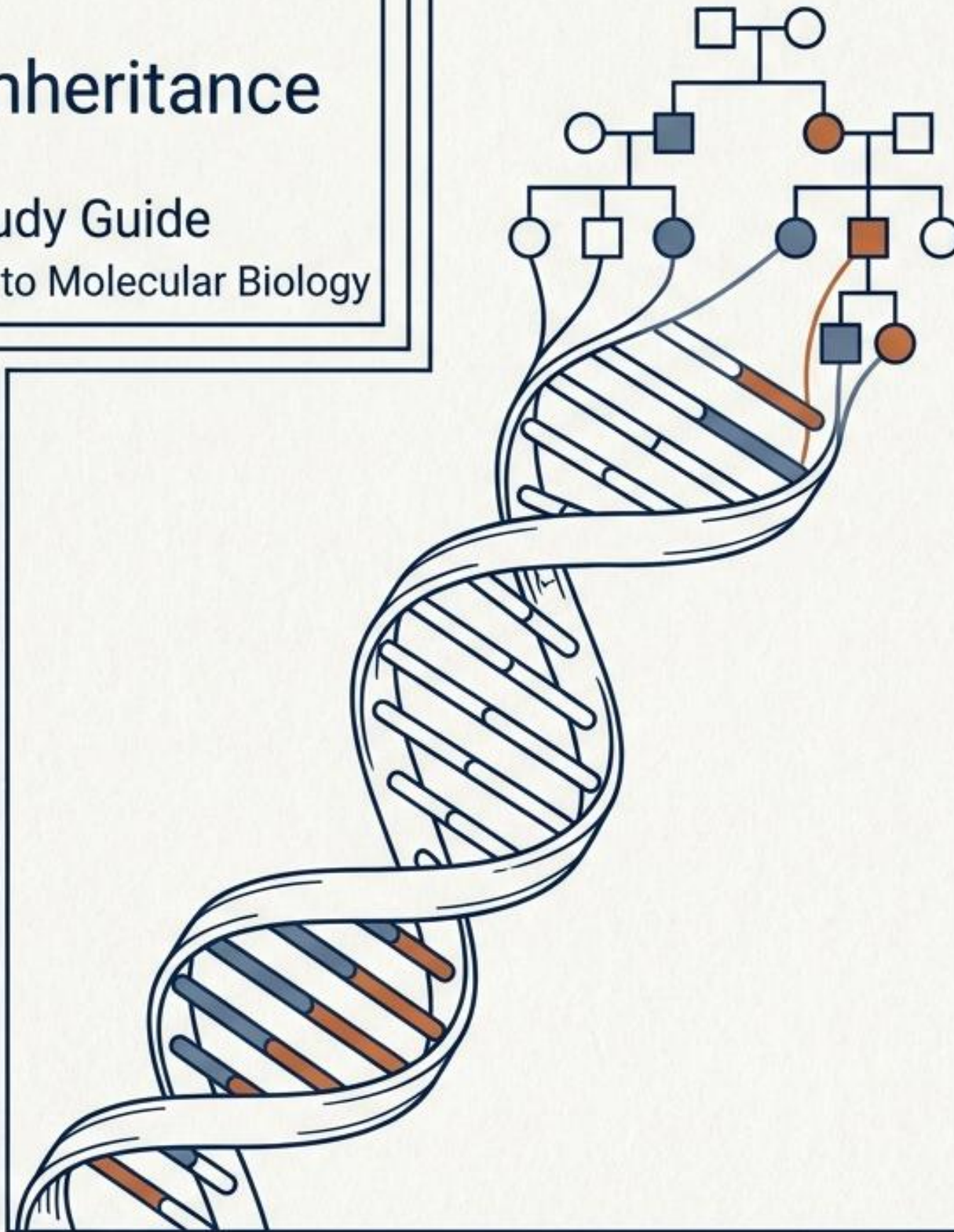


Unit 3 Mastery: The Architectre of Inheritance

A Comprehensive Visual Study Guide

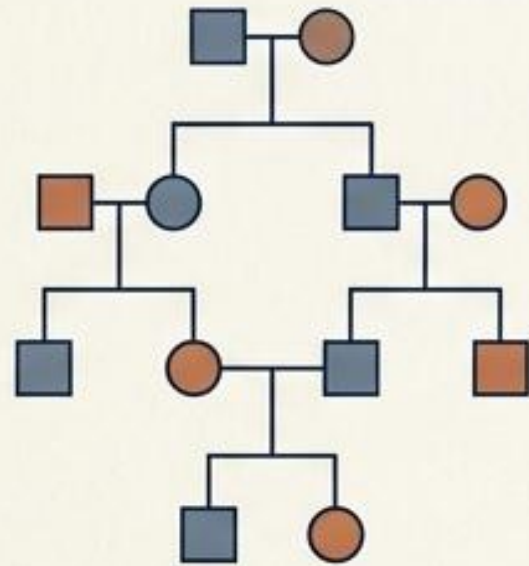
Chapters 5 & 6: Classical Genetics to Molecular Biology



The Scale of Heredity: Tracing Traits from Organism to Atom

Act I: Macro

Visible patterns, pedigrees, and probabilities.

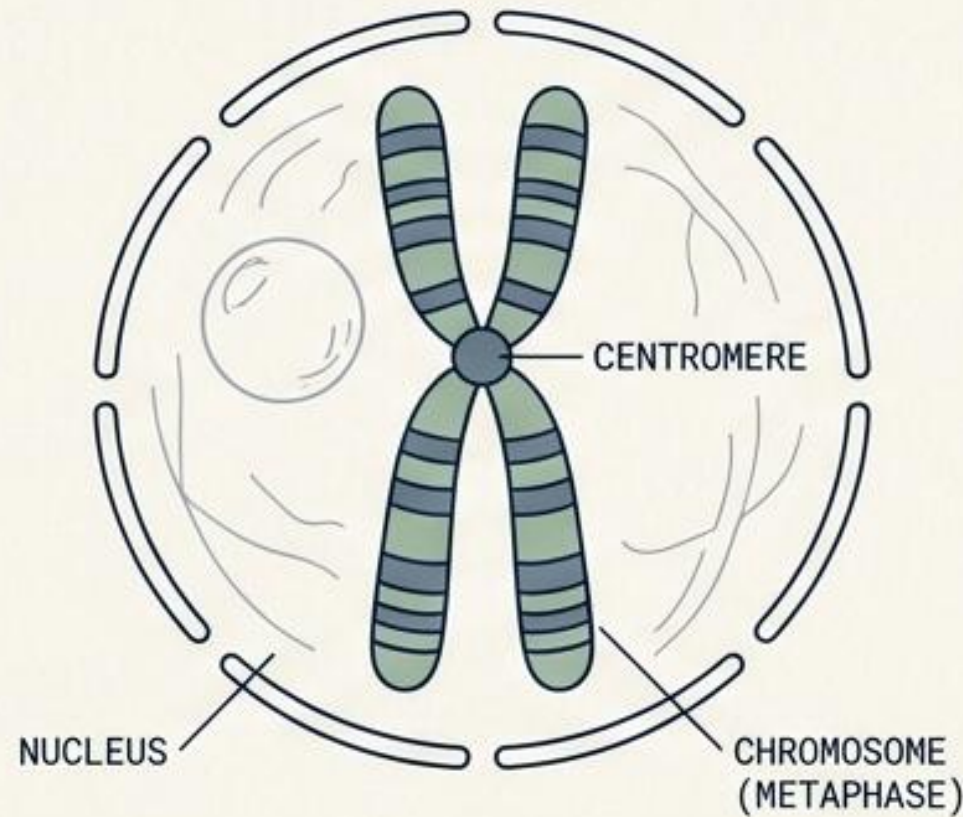


■ Slate Blue: Affected
■ Burnt Orange: Unaffected

MODEL: HOMO SAPIENS
OBSERVED PATTERN: DOMINANT INHERITANCE

Act II: Meso

Meiosis, karyotypes, and chromosomal sorting.

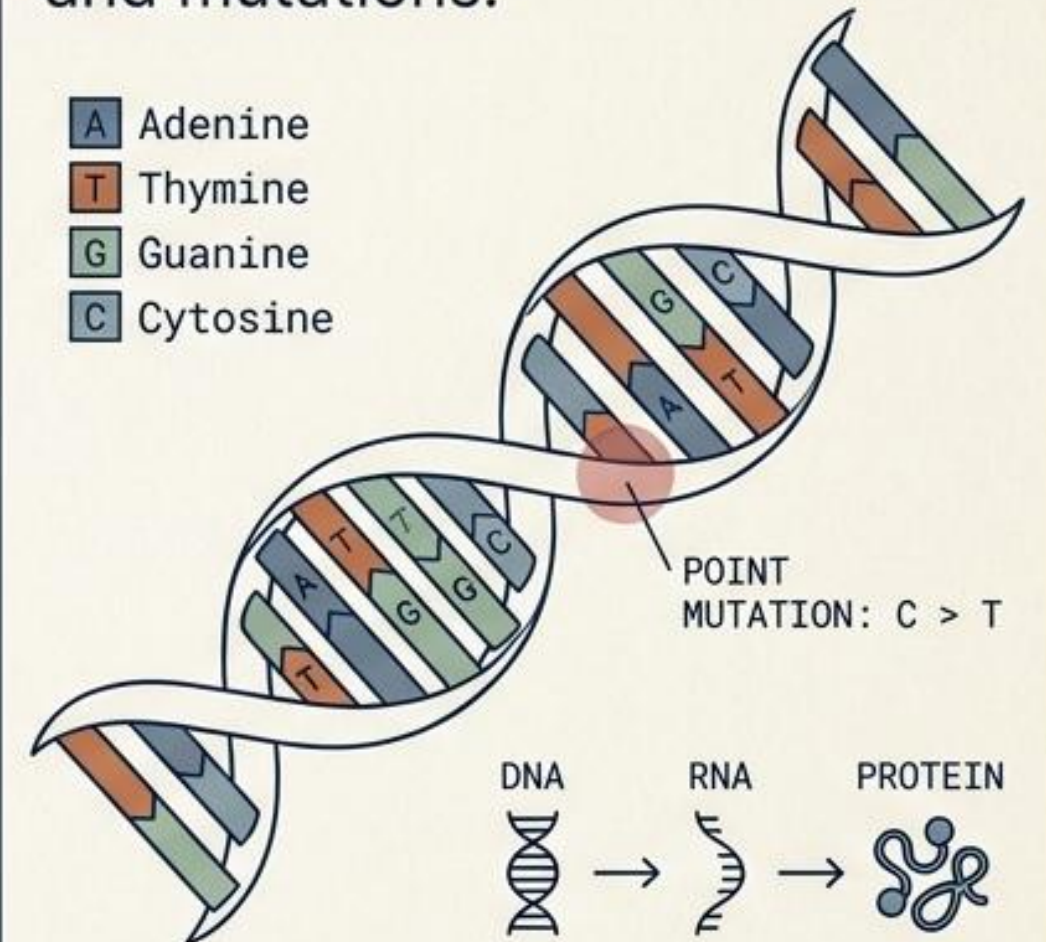


SCALE: MICROSCOPIC
KARYOTYPE: 23 PAIRS
PROCESS: MEIOSIS I & II

Act III: Micro

Chemical codes, Central Dogma, and mutations.

A Adenine
T Thymine
G Guanine
C Cytosine



UNIT: BASE PAIR
CODE: GENETIC SEQUENCE
CENTRAL DOGMA PATHWAY

Reading the Past: The Pedigree Diagnostic Flowchart

Pedigree Symbol Key



Square = Male



Circle = Female



Shaded = Expresses Trait



Horizontal Line = Mating



Vertical Line = Offspring

GENERATION I



Black Male
BB or Bb



Red Female
bb

GENERATION II



Black Female



Black Male



Black Male



Red Female

GENERATION III



Red Female

Diagnostic Check

Question: Does the trait skip a generation?

Observation: Two dominant phenotypes produced a recessive offspring.

Conclusion: It is a Recessive Trait. The hidden 'b' alleles emerged from heterozygous 'Bb' parents.

Decoding the Organism: Genotypes, Phenotypes, and Probability

Genotype: The Invisible Code

The specific combination of inherited alleles.

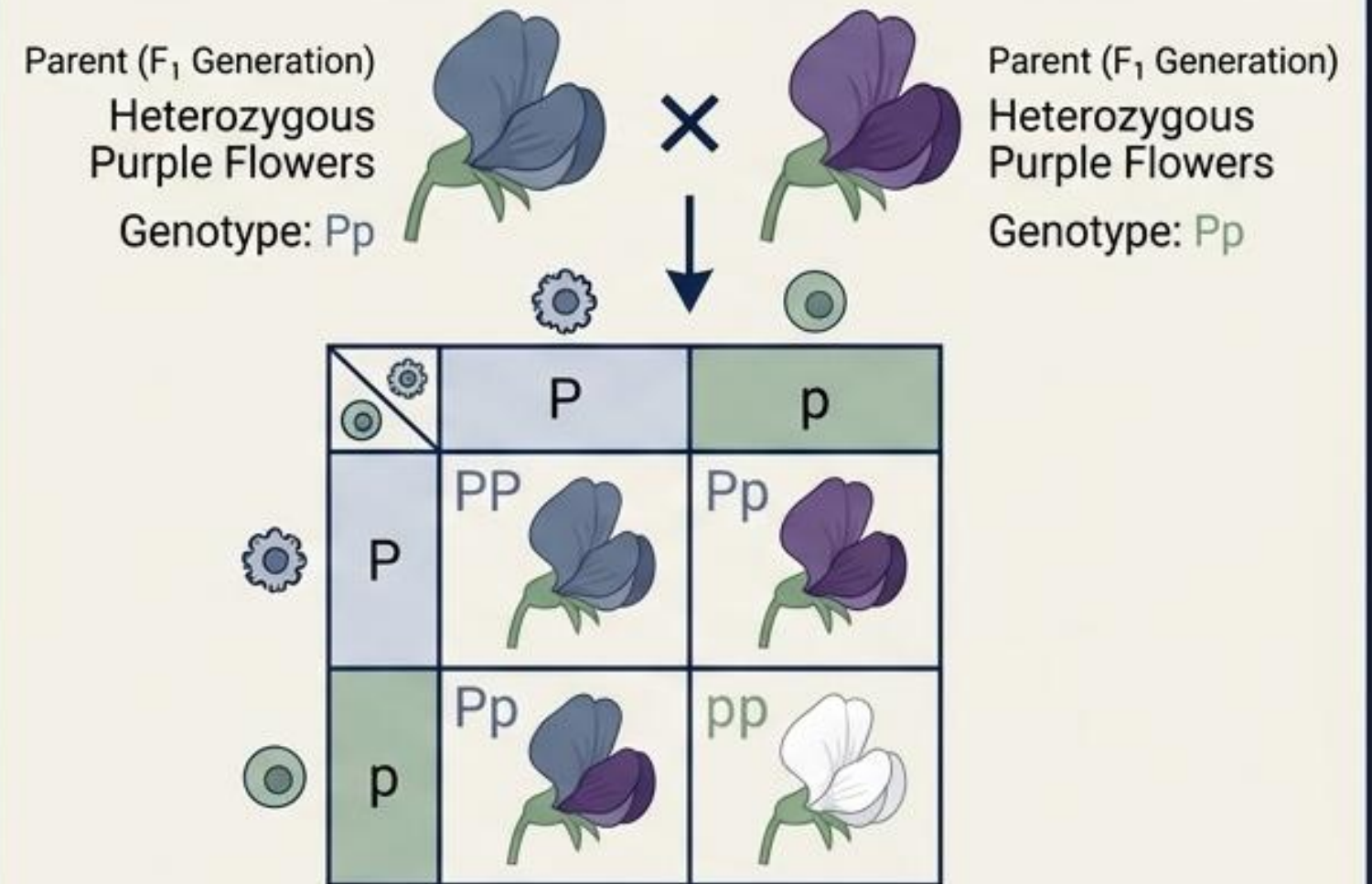
- Homozygous Dominant (PP)
- Heterozygous (Pp)
- Homozygous Recessive (pp)

Phenotype: The Physical Expression

The observable physical trait driven by the genotype.

- Example: Purple Flowers or White Flowers

F₁ Cross (Pp x Pp)



The F₂ Generation Math:
Genotypic Ratio = 1:2:1 (PP : Pp : pp)
Phenotypic Ratio = 3 Purple : 1 White (75% Probability)

Beyond Simple Dominance: The Complex Inheritance Matrix

Pattern	The Rule	Classic Example
Incomplete Dominance	Neither allele dominates; traits blend into a new phenotype.	Red + White = Pink Flowers 
Codominance	Both alleles are expressed equally and distinctly.	Black + White = Checkered Chicken Feathers 
Multiple Alleles	A single gene has more than two possible alleles in the population.	Human Blood Types (A, B, O alleles) 
Sex-Linked	Genes located on the X or Y chromosome; XY males show higher risk.	Colorblindness 
Polygenic	A single trait regulated by many interacting genes.	Continuous spectrum of human height 

Tracking Multiple Traits: Independent Assortment

F₁ Cross (SsPp x SsPp)

	SP	Sp	sP	sp
SP	 SSPP  Smooth & Purple	 SSPp  Smooth & Purple	 SsPp  Smooth & Purple	 SsPp  Smooth & Purple
Sp	 SSPp  Smooth & Purple	 SSpp  Smooth & White	 SsPp  Smooth & Purple	 Sspp  Smooth & White
sP	 SSPp  Smooth & Purple	 SsPp  Smooth & Purple	 ssPp  Wrinkled & Purple	 sspp  Wrinkled & White
sp	 SSPp  Smooth & Purple	 Sspp  Smooth & White	 ssPp  Wrinkled & Purple	 sspp  Wrinkled & White

The 9:3:3:1 Phenotypic Ratio

-  9 = Smooth & Purple
-  3 = Smooth & White
-  3 = Wrinkled & Purple
-  1 = Wrinkled & White

Mendel's Law of Independent Assortment

Inheriting a dominant seed shape (Smooth) does not force the inheritance of a dominant flower color (Purple). Traits separate independently.

The Ultimate Combination: Human ABO Blood Typing

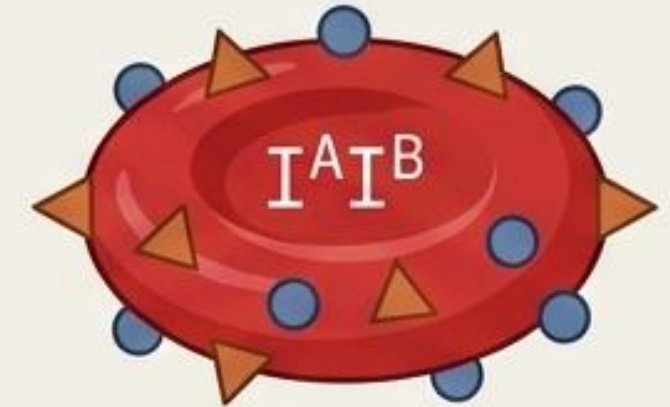
Rule 1: Multiple Alleles

Instead of two, three distinct alleles exist in the human population pool:



Rule 2: Codominance

The I^A and I^B alleles are equally dominant. Inheriting both results in both being expressed.



Type AB Blood

Rule 3: Simple Dominance (ABO & Rh Factor)

ABO System:

Both I^A and I^B mask the recessive "i".

Type O blood requires the strict "ii" genotype.



Type O Blood
ii

$I^A i \rightarrow$ Type A

$I^B i \rightarrow$ Type B

Rh System:

The positive (+) or negative (-) factor is controlled by a separate gene via simple dominance.

Having the Rh protein (R) dominates lacking it (r).



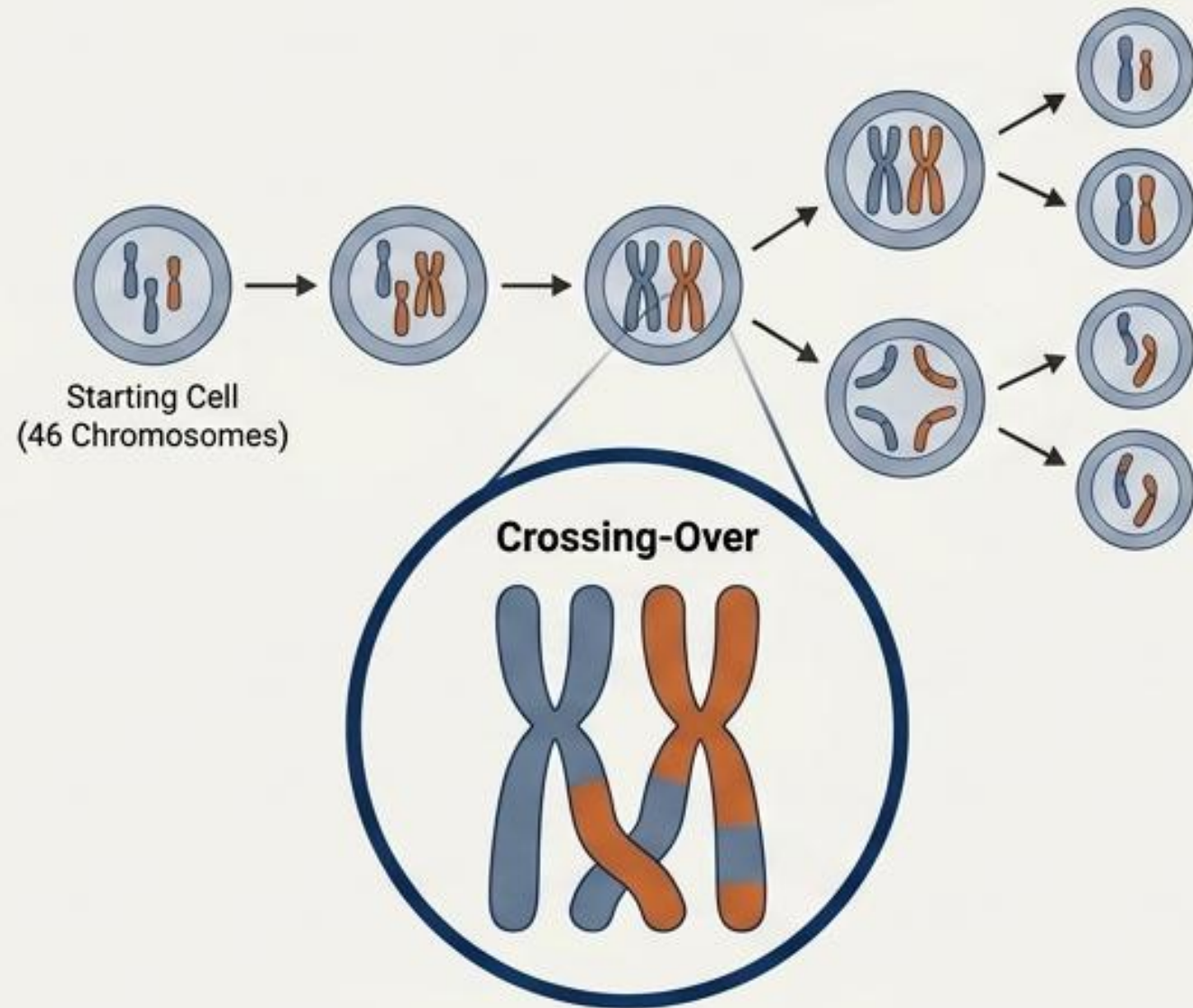
Rh+
RR or Rr



Rh-
rr

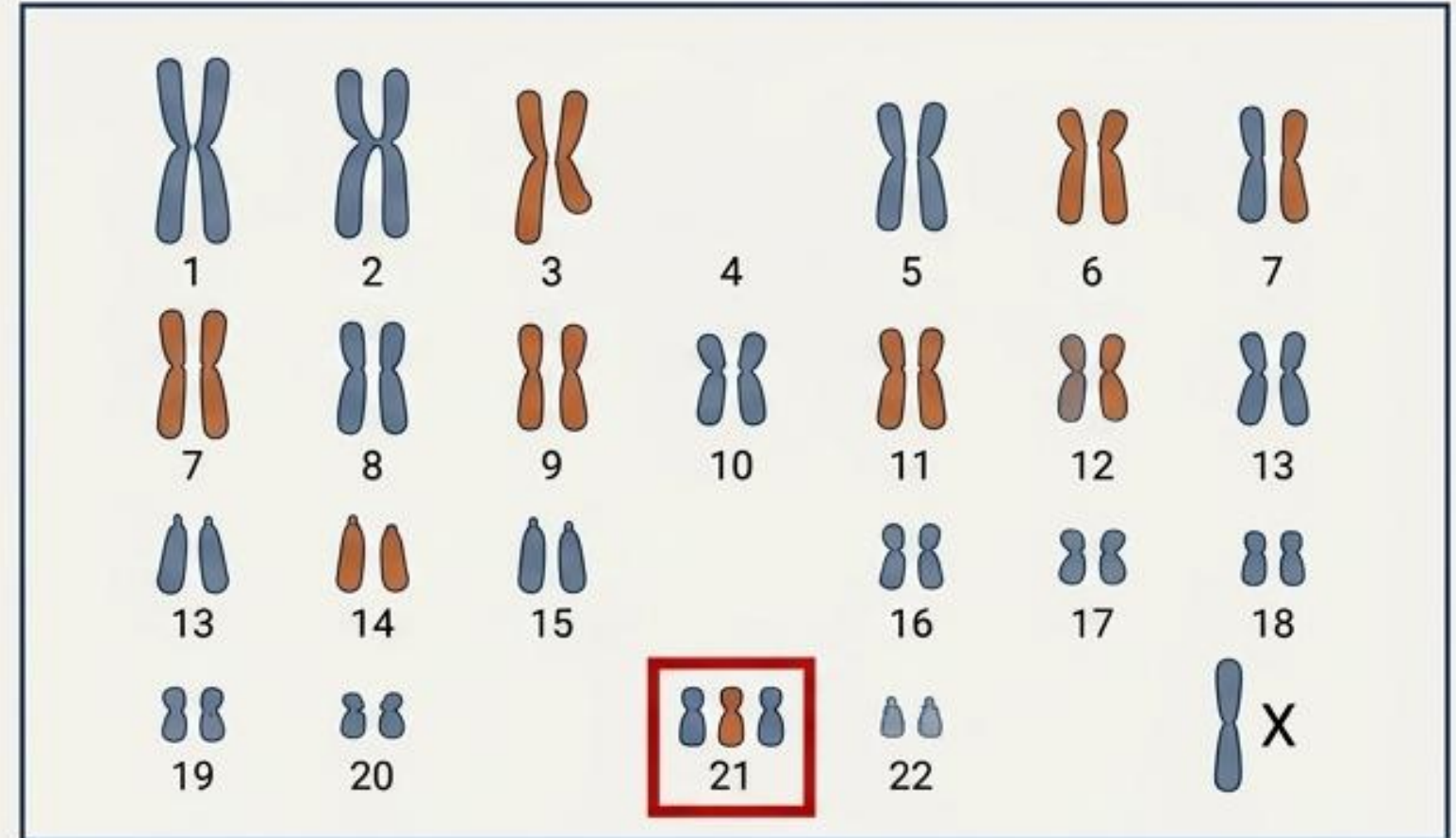
Vehicles of Inheritance: Chromosomes, Meiosis, and Nondisjunction

Meiosis & Crossing-Over



Meiosis halves the chromosome count for reproduction, while **crossing-over** guarantees unique genetic variations.

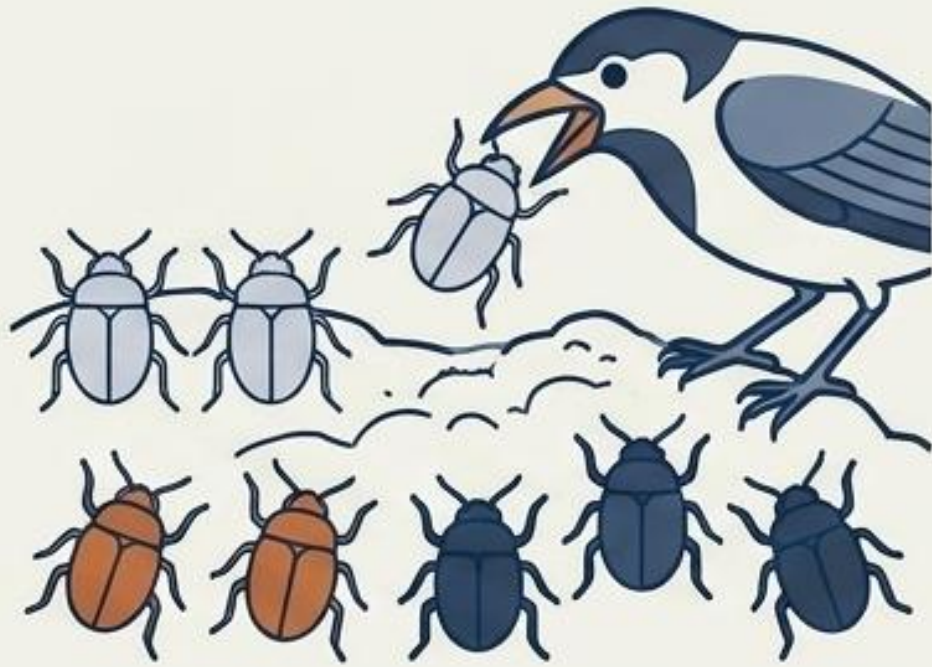
Clinical Karyogram



Disorder	Mechanism	Notation
Down Syndrome	Trisomy 21	47, XX, +21
Turner Syndrome	Monosomy X	45, X0
Klinefelter Syndrome	Extra X	47, XXY

The Macro-Scale: Forces of Population Genetics

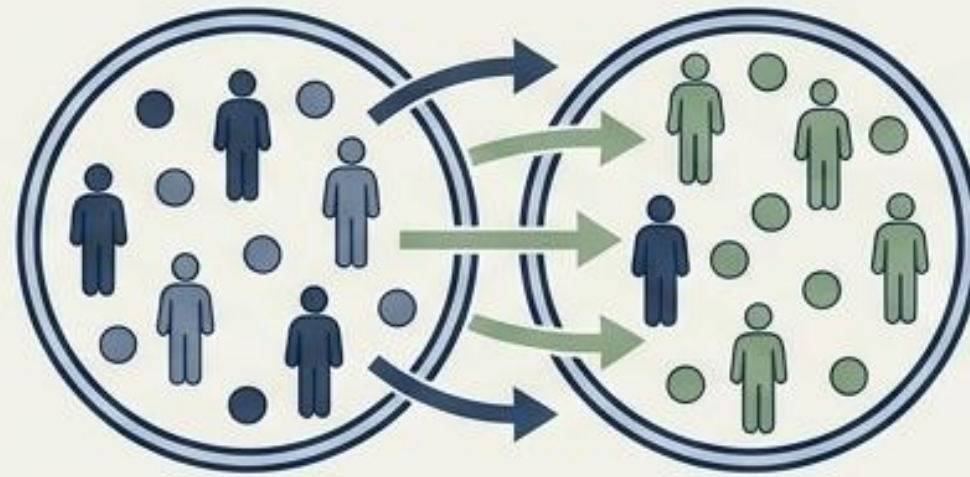
Natural Selection



Environmental pressure

Advantageous alleles survive, reproduce, and increase in frequency over time.

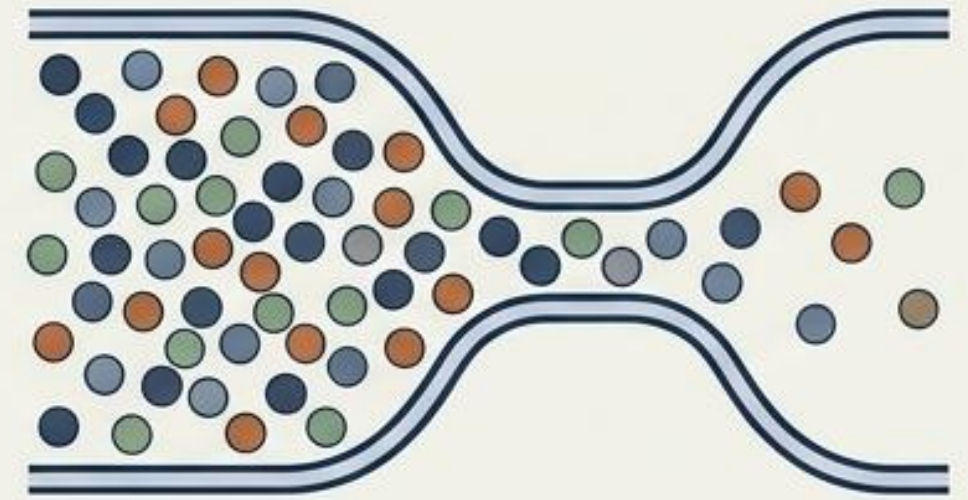
Gene Flow



Migration

The physical movement of alleles between groups keeps adjacent populations genetically similar.

Genetic Drift



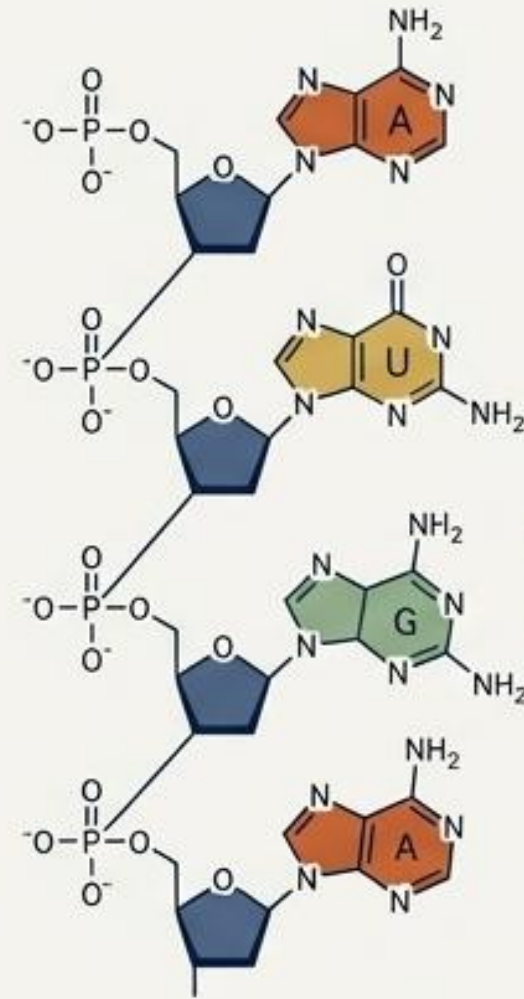
Chance events

Random disasters or bottleneck effects drastically and unpredictably alter allele frequencies, especially in small populations.

The Chemical Code: Deconstructing DNA and RNA

DNA Vault

- Structure: Double-stranded Helix
- Sugar: Deoxyribose
- Bases: Adenine (A), Cytosine (C), Guanine (G), Thymine (T)
- Function: Permanent long-term storage locked inside the nucleus.



RNA
(Ribonucleic Acid)

Vs



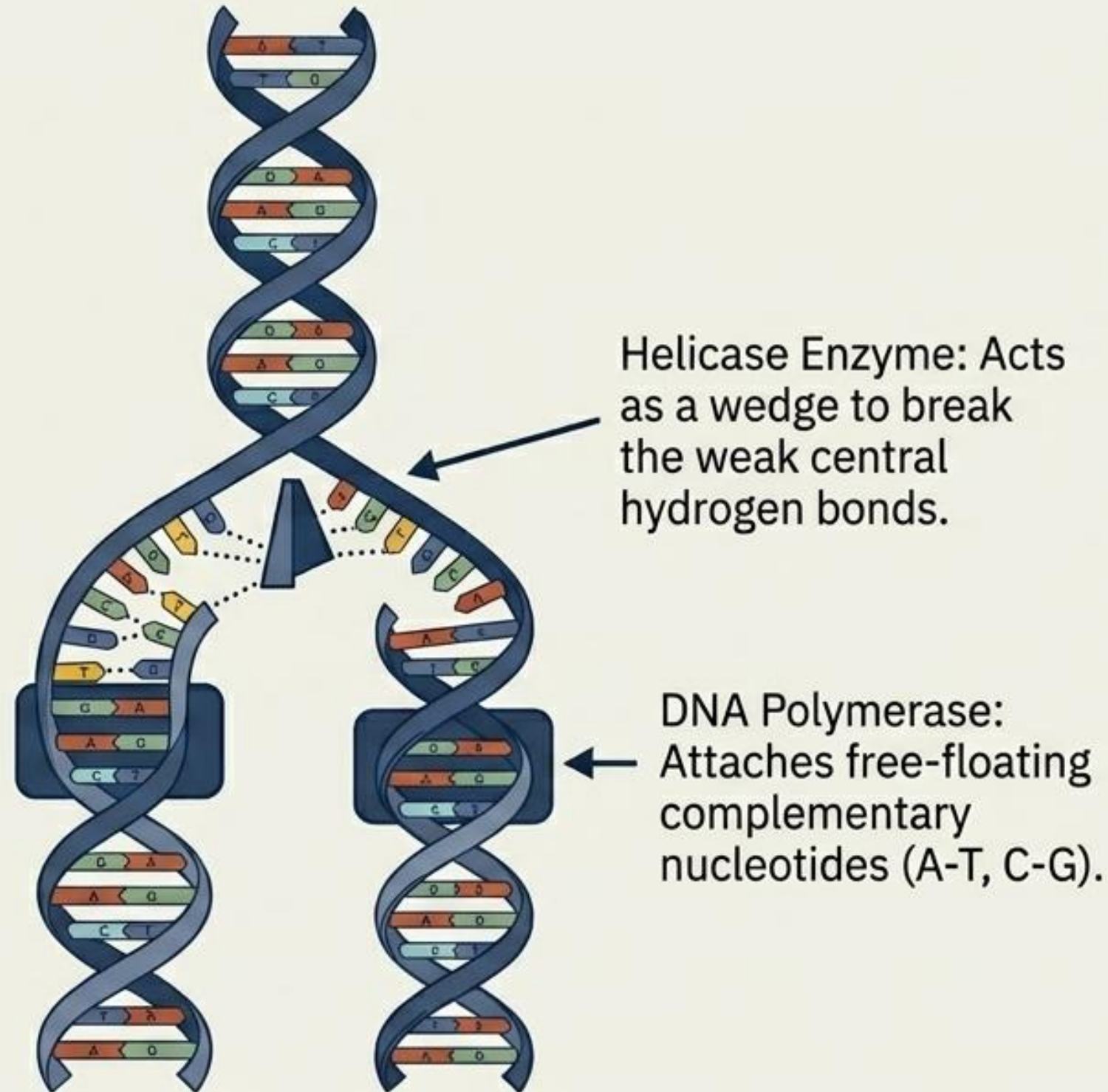
DNA
(Deoxyribonucleic Acid)

RNA Messenger

- Structure: Single-stranded
- Sugar: Ribose
- Bases: Adenine (A), Cytosine (C), Guanine (G), Uracil (U)
- Function: Short-term transmission; moves freely to the cytoplasm.

Chargaff's Base-Pairing Rules: Adenine strictly pairs with Thymine (or Uracil in RNA).
Cytosine strictly pairs with Guanine.

Copying the Manual: Semi-Conservative DNA Replication



The Semi-Conservative Rule

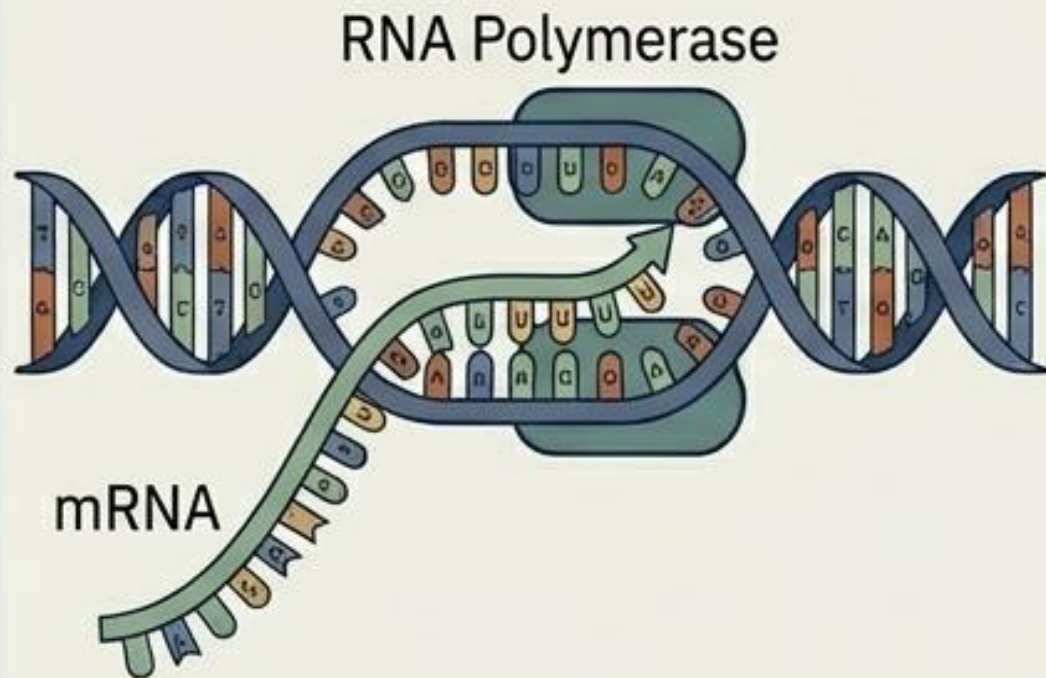
Every newly built DNA double helix consists of exactly one original parent strand (ensuring protection accuracy) and one newly synthesized matching strand.

The Central Dogma: Information to Architecture

DNA → RNA → Protein

The Nucleus Vault

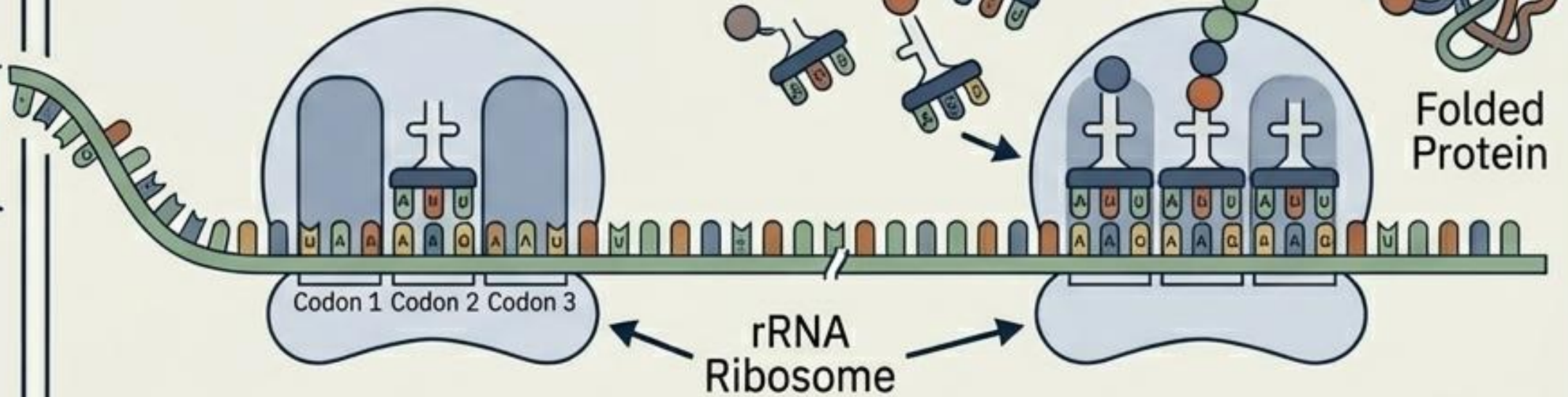
1. Transcription



RNA Polymerase transcribes a temporary mRNA copy of the gene, swapping Thymine for Uracil.

The Cytoplasm Factory

2. Translation



The ribosome reads the mRNA sequence three letters at a time (a Codon).

tRNA delivery vehicles match codons and drop off specific amino acids to build a folded 3D polypeptide chain (Protein).

System Errors: The Mutation Spectrum

Original Base Sequence:

THE FAT CAT ATE THE RAT

Category 1: Point Mutations (Substitutions)

Silent

Base changes, but redundancy results in the exact same amino acid. (Invisible effect).

Missense

Changes the amino acid.
E.g., THE FAT **HAT** ATE THE RAT
(Slightly altered function).

Nonsense

Accidentally creates a STOP codon. Protein is cut short and ruined.

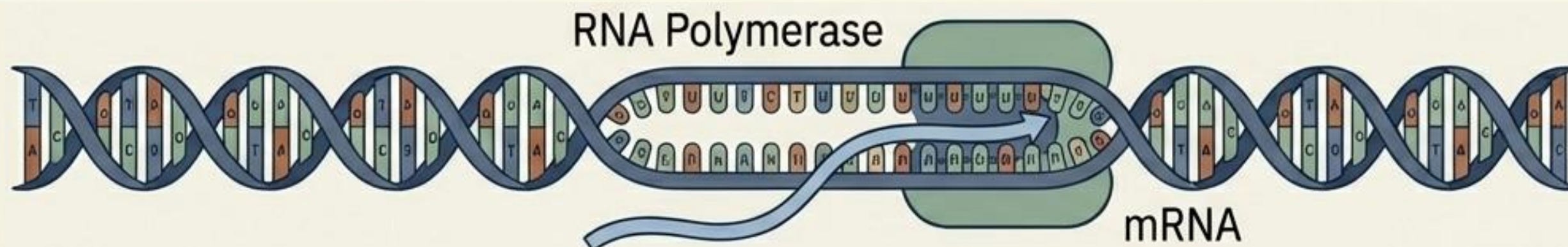
Category 2: Frameshift Mutations (Insertions/Deletions)

Deleting or inserting a single letter shifts the entire reading frame, causing catastrophic functional failure.

THE FAT ~~C~~AT ATE THE RAT
THE FAT ATA TET HER AT

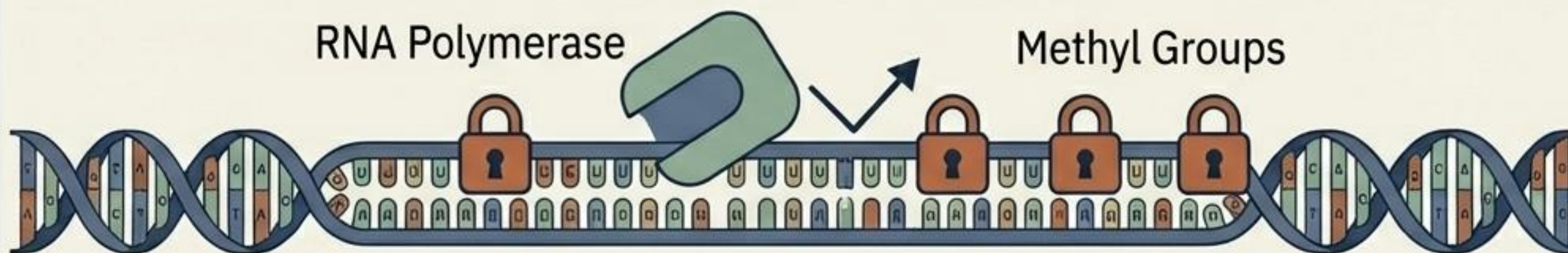
Epigenetics: The Software Managing the Hardware

Active Gene



Every cell shares the identical DNA hardware.
Without obstruction, the gene is transcribed (turned ON).

Silenced Gene (DNA Methylation)



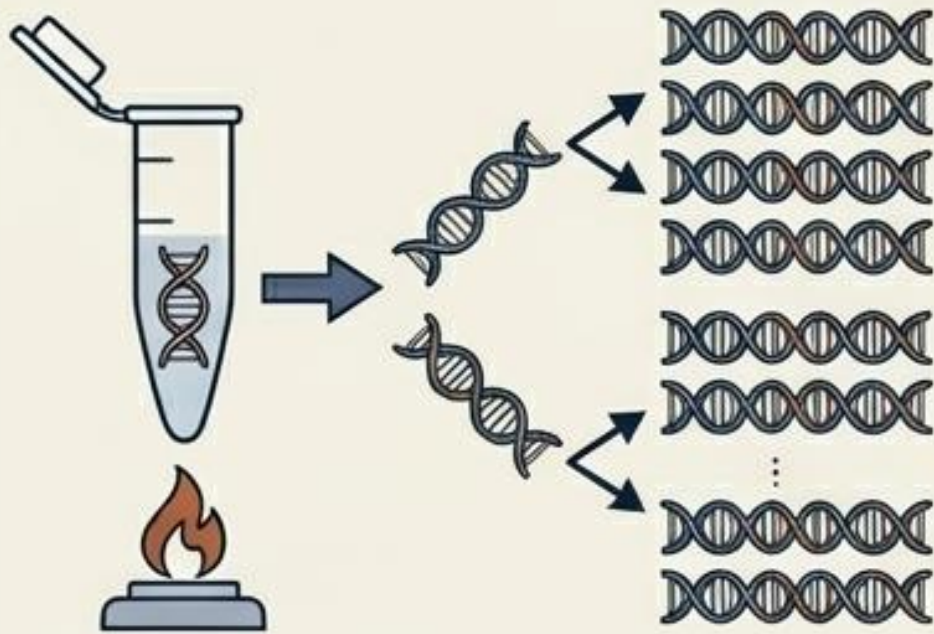
Chemical tags physically block the reading mechanism.
The gene is silenced (turned OFF) without altering the underlying sequence.

Environmental Triggers

- Chronic Stress
- Diet & Toxins
- Historical/Intergenerational Trauma

The Molecular Toolbox: Manipulating the Code

1. Copying (PCR)



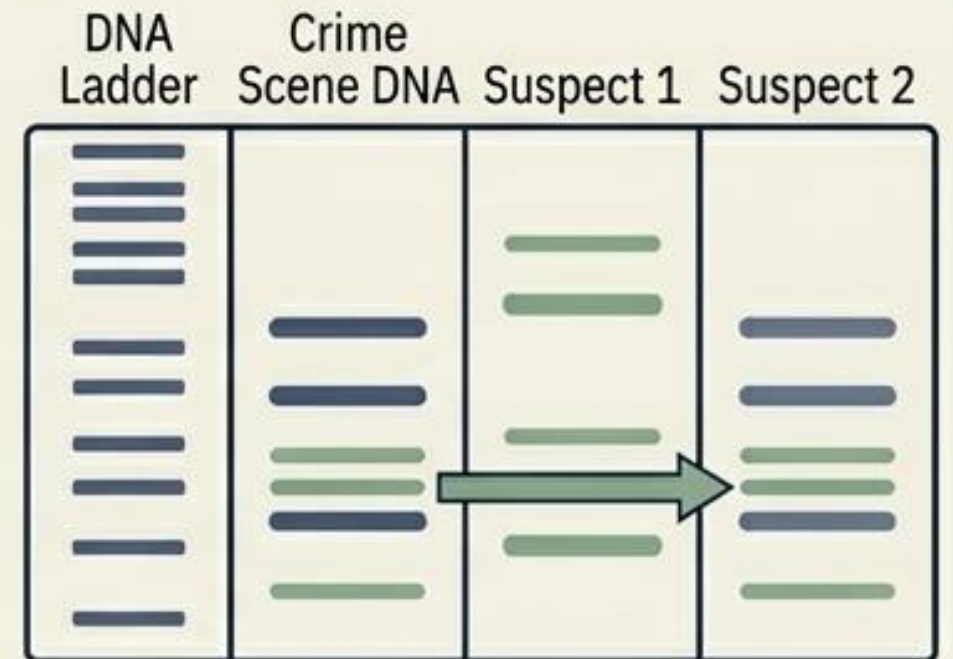
Polymerase Chain Reaction: Rapid heating and cooling forces heat-resistant DNA polymerase to amplify a tiny sample into billions of copies.

2. Cutting (Restriction Enzymes)



Enzymes act as molecular scissors, scanning the DNA and cutting the strands only at highly specific letter target sequences.

3. Sorting (Gel Electrophoresis)



An electrical current pulls negatively charged DNA fragments through an agarose gel sponge. Smaller pieces travel faster, creating a distinct barcode.